

Jungho Lee

RESEARCH SCIENTIST AT NAVER AI LAB

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RESEARCH INTERESTS

3D Computer Vision

- Geometric Foundation Models
- 3D Scene Representation
- 3D Neural Rendering

World Models

- Video World Models

EDUCATION

Yonsei University

INTERATED M.S./PH.D IN ELECTRICAL AND ELECTRONIC ENGINEERING

- Image and Video Pattern Recognition Lab.
- Advisor: Prof. Sangyoun Lee

Seoul, South Korea

Sep. 2021 - Aug. 2026

Yonsei University

B.S. IN ELECTRICAL & ELECTRONIC ENGINEERING

- 2-Year Military Service (2017-2019)

Seoul, South Korea

Mar. 2015 - Aug. 2021

Whimoon High School

Seoul, South Korea

Mar. 2010 - Feb. 2013

EXPERIENCE

NAVER AI Lab

RESEARCH SCIENTIST

- Generation Research

Seongnam, South Korea

Aug. 2026 - Present

NAVER Cloud

RESEARCH INTERN

- 3D Scene Representation from Defocused Images
- 3D Human Avatar Generation
- Mentor: Ho-Deok Jang

Seongnam, South Korea

Aug. 2024 - Feb. 2025

Military Service

REPUBLIC OF KOREA AIR FORCE

- Bird Alert Team (BAT) & Foreign Object Debris (FOD) Crew

Seongnam, South Korea

Sep. 2017 - Aug. 2019

PUBLICATIONS

First Author

[F7] DensiTok: Making Feed-Forward 3D Gaussian Splatting See More Views Than It Is Given

2027

MINHYEOK LEE*, JUNGHOO LEE*, MINSEOK KANG, HEESEUNG CHOI, IG-JAE KIM, SANGYOUN LEE

Submitted to International Conference on Learning Representations (ICLR)

[F6] SwiftVGGT: A Scalable Visual Geometry Grounded Transformer for Large-Scale Scenes

2026

JUNGHOO LEE, MINHYEOK LEE, SUNGHUN YANG, MINSEOK KANG, SANGYOUN LEE

The IEEE/CVF Computer Vision and Pattern Recognition (CVPR) Findings

[F5] CoMoGaussian: Continuous Motion-Aware Gaussian Splatting from Motion-Blurred Images

2025

JUNGHOO LEE, DONGHYEONG KIM, DOGYOON LEE, SUHWAN CHO, MINHYEOK LEE, WONJOON LEE, TAEHO KIM, DONGYOON WEE, SANGYOUN LEE

The IEEE/CVF International Conference on Computer Vision (ICCV)

[F4] CoCoGaussian: Leveraging Circle of Confusion for Gaussian Splatting from Defocused Images	2025
JUNGHO LEE, SUHWAN CHO, TAEHO KIM, HO-DEOK JANG, MINHYEOK LEE, GEONHO CHA, DONGYOON WEE, DOGYOON LEE, SANGYOUN LEE <i>The IEEE/CVF Computer Vision and Pattern Recognition (CVPR)</i>	
[F3] SMURF: Continuous Dynamics for Motion-Deblurring Radiance Fields	2025
JUNGHO LEE, DOGYOON LEE, MINHYEOK LEE, DONGHYEONG KIM, SANGYOUN LEE <i>2nd Workshop on Neural Fields Beyond Conventional Cameras (CVPRW)</i>	
[F2] Hierarchically Decomposed Graph Convolutional Networks for Skeleton-Based Action Recognition	2023
JUNGHO LEE, MINHYEOK LEE, DOGYOON LEE, SANGYOUN LEE <i>The IEEE/CVF International Conference on Computer Vision (ICCV)</i>	
[F1] Leveraging Spatio-Temporal Dependency for Skeleton-Based Action Recognition	2023
JUNGHO LEE, MINHYEOK LEE, SUHWAN CHO, SUNGMIN WOO, SUNGJUN JANG, SANGYOUN LEE <i>The IEEE/CVF International Conference on Computer Vision (ICCV)</i>	
Conference Proceedings	
[C13] Revisiting Weakly-Supervised Video Scene Graph Generation via Pair Affinity Learning	2026
MINSEOK KANG, MINHYEOK LEE, MINJUNG KIM, JUNGHO LEE, DONGHYEONG KIM, SUNGMIN WOO, INSEOK JEON, SANGYOUN LEE <i>European Conference on Computer Vision (ECCV)</i>	
[C12] MoRGs: Efficient Per-Gaussian Motion Reasoning for Streamable Dynamic 3D Scenes	2026
WONJOON LEE, SUNGMIN WOO, DONGHYEONG KIM, JUNGHO LEE, SANGHEON PARK, SANGYOUN LEE <i>Proceedings of the IEEE/CVF Computer Vision and Pattern Recognition (CVPR)</i>	
[C11] SwiftVGGT: A Scalable Visual Geometry Grounded Transformer for Large-Scale Scenes	2026
JUNGHO LEE, MINHYEOK LEE, SUNGHUN YANG, MINSEOK KANG, SANGYOUN LEE <i>Proceedings of the IEEE/CVF Computer Vision and Pattern Recognition (CVPR) Findings</i>	
[C10] MonoCLUE: Object-Aware Clustering Enhances Monocular 3D Object Detection	2026
SUNGHUN YANG, MINHYEOK LEE, JUNGHO LEE, SANGYOUN LEE <i>The Association for the Advancement of Artificial Intelligence (AAAI)</i>	
[C9] CoMoGaussian: Continuous Motion-Aware Gaussian Splatting from Motion-Blurred Images	2025
JUNGHO LEE, DONGHYEONG KIM, DOGYOON LEE, SUHWAN CHO, MINHYEOK LEE, WONJOON LEE, TAEHO KIM, DONGYOON WEE, SANGYOUN LEE <i>Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)</i>	
[C8] CMTM: Cross-Modal Token Modulation for Unsupervised Video Object Segmentation	2025
INSEOK JEON, SUHWAN CHO, MINHYEOK LEE, SEUNGHOON LEE, MINSEOK KANG, JUNGHO LEE, CHAEWON PARK, DONGHYEONG KIM, SANGYOUN LEE <i>IEEE International Conference on Image Processing (ICIP)</i>	
[C7] CoCoGaussian: Leveraging Circle of Confusion for Gaussian Splatting from Defocused Images	2025
JUNGHO LEE, SUHWAN CHO, TAEHO KIM, HO-DEOK JANG, MINHYEOK LEE, GEONHO CHA, DONGYOON WEE, DOGYOON LEE, SANGYOUN LEE <i>Proceedings of the IEEE/CVF Computer Vision and Pattern Recognition (CVPR)</i>	
[C6] Effective SAM Combination for Open-Vocabulary Semantic Segmentation	2025
MINHYEOK LEE, SUHWAN CHO, JUNGHO LEE, SUNGHUN YANG, HEESEUNG CHOI, IG-JAE KIM, SANGYOUN LEE <i>Proceedings of the IEEE/CVF Computer Vision and Pattern Recognition (CVPR) - Oral Presentation</i>	
[C5] Video Diffusion Models are Strong Video Inpainter	2025
MINHYEOK LEE, SUHWAN CHO, CHAJIN SHIN, JUNGHO LEE, SUNGHUN YANG, SANGYOUN LEE <i>The Association for the Advancement of Artificial Intelligence (AAAI)</i>	
[C4] Guided Slot Attention for Unsupervised Video Object Segmentation	2024
MINHYEOK LEE, DOGYOON LEE, SUHWAN CHO, CHAEWON PARK, JUNGHO LEE, SANGYOUN LEE <i>Proceedings of the IEEE/CVF Computer Vision and Pattern Recognition (CVPR)</i>	

[C3] Hierarchically Decomposed Graph Convolutional Networks for Skeleton-Based Action Recognition JUNGHOO LEE, MINHYEOK LEE, DOGYOON LEE, SANGYOUN LEE <i>Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)</i>	2023
[C2] Leveraging Spatio-Temporal Dependency for Skeleton-Based Action Recognition JUNGHOO LEE, MINHYEOK LEE, SUHWAN CHO, SUNGMIN WOO, SUNGJUN JANG, SANGYOUN LEE <i>Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)</i>	2023
[C1] Detection-Identification Balancing Margin Loss for One-Stage Multi-Object Tracking HEANSUNG LEE, SUHWAN CHO, SUNGJUN JANG, JUNGHOO LEE, SANGYOUN LEE <i>IEEE International Conference on Image Processing (ICIP)</i>	2022

Journals

[J5] Synchronizing Vision and Language: Bidirectional Token-Masking AutoEncoder for Referring Image Segmentation MINHYEOK LEE, DOGYOON LEE, JUNGHOO LEE, SUHWAN CHO, SANGYOUN LEE <i>Pattern Recognition (PR)</i>	2026
[J4] GenCLIP: Generalizing CLIP Prompts for Zero-Shot Anomaly Detection DONGHYEONG KIM, CHAEWON PARK, SUHWAN CHO, HYEONJEONG LIM, MINSEOK KANG, JUNGHOO LEE, SANGYOUN LEE <i>Pattern Recognition (PR)</i>	2025
[J3] Sparse-DeRF: Deblurred Neural Radiance Fields from Sparse View DOGYOON LEE, DONGHYEONG KIM, JUNGHOO LEE, MINHYEOK LEE, SEUNGHOON LEE, SANGYOUN LEE <i>IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)</i>	2025
[J2] Treating Motion as Option with Output Selection for Unsupervised Video Object Segmentation SUHWAN CHO, MINHYEOK LEE, JUNGHOO LEE, MYEONGAH CHO, SANGYOUN LEE <i>IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)</i>	2025
[J1] Multi-Scale Structural Graph Convolutional Network for Skeleton-Based Action Recognition SUNGJUN JANG, HEANSUNG LEE, WOJIN KIM, JUNGHOO LEE, SUNGMIN WOO, SANGYOUN LEE <i>IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)</i>	2024

Workshops

[W5] GenCLIP: Generalizing CLIP Prompts for Zero-Shot Anomaly Detection DONGHYEONG KIM, CHAEWON PARK, SUHWAN CHO, HYEONJEONG LIM, MINSEOK KANG, JUNGHOO LEE, SANGYOUN LEE <i>The 3rd Workshop in Vision-based Industrial Inspection (ICCVW)</i>	2025
[W4] Find First, Track Next: Decoupling Identification and Propagation in Referring Video Object Segmentation SUHWAN CHO, SEUNGHOON LEE, MINHYEOK LEE, JUNGHOO LEE, SANGYOUN LEE <i>The 7th Large-Scale Object Segmentation (LSVOS) Workshop (ICCVW)</i>	2025
[W3] DepthFlow: Exploiting Depth-Flow Structural Correlations for Unsupervised Video Object Segmentation SUHWAN CHO, MINHYEOK LEE, JUNGHOO LEE, DONGHYEONG KIM, SANGYOUN LEE <i>The 7th Large-Scale Object Segmentation (LSVOS) Workshop (ICCVW)</i>	2025
[W2] TransFlow: Motion Knowledge Transfer from Video Diffusion Models to Video Salient Object Detection SUHWAN CHO, MINHYEOK LEE, JUNGHOO LEE, SUNGHUN YANG, SANGYOUN LEE <i>The 7th Large-Scale Object Segmentation (LSVOS) Workshop (ICCVW)</i>	2025
[W1] SMURF: Continuous Dynamics for Motion-Deblurring Radiance Fields JUNGHOO LEE, DOGYOON LEE, MINHYEOK LEE, DONGHYEONG KIM, SANGYOUN LEE <i>2nd Workshop on Neural Fields Beyond Conventional Cameras (CVPRW)</i>	2025

Preprints

DensiTok: Making Feed-Forward 3D Gaussian Splatting See More Views Than It Is Given MINHYEOK LEE*, JUNGHO LEE* , MINSEOK KANG, HEESEUNG CHOI, IG-JAE KIM, SANGYOUN LEE <i>Submitted to International Conference on Learning Representations (ICLR)</i>	2027
OTT-Vid: Optimal Transport Temporal Token Compression for Video Large Language Models MINSEOK KANG, MINHYEOK LEE, JUNGHO LEE , MINJUNG KIM, DONGHYEONG KIM, DAYEON LEE, HEESEUNG CHOI, IG-JAE KIM, SANGYOUN LEE <i>Submitted to International Conference on Learning Representations (ICLR)</i>	2027
What is the Event and When does it Happen? INSEOK JEON, SUHWAN CHO, SEUNGHOON LEE, JUNGHO LEE , SANGYOUN LEE <i>Submitted to International Conference on Learning Representations (ICLR)</i>	2027
SCALE: Correcting Scale Drift with Contamination-Agnostic Loop Closure in Geometric Foundation-Model SLAM CHUNGJAE CHOE, JUNGHO LEE , SEOKJUN LEE, NAKMYOUNG SUNG, CHANGICK KIM <i>Submitted to The Association for the Advancement of Artificial Intelligence (AAAI)</i>	2026
Georeferenced Gaussian Splatting with RTK-GPS-Constrained Camera Pose Refinement for 3D Reconstruction CHEOLHWAN KIM, WONJUN CHOI, YOUNGMOK KWON, JUNGHO LEE , MINHYEOK LEE, HONG-GYOO SOHN <i>Submitted to Automation in Construction (TGRS)</i>	2026
ESC-Net++: Effective SAM Combination for Open-Vocabulary Image and Video Segmentation MINHYEOK LEE, JUNGHO LEE , SUHWAN CHO, SUNGHUN YANG, SANGYOUN LEE <i>Submitted to Transactions on Pattern Analysis and Machine Intelligence (TPAMI)</i>	2026
STATIC: Surface-normal-guided Two-branch Alignment and Temporal Integration for Video Depth Estimation SUNGHUN YANG, MINHYEOK LEE, SUHWAN CHO, JUNGHO LEE , SANGYOUN LEE <i>Submitted to Pattern Recognition (PR)</i>	2026

PATENTS

Domestic Patent

- [P3] Palmprint Recognition Method. KR-Application No.10-2023-0156996, Nov., 2023.
- [P2] Skeleton Graph-based Action Recognition Device and Method. KR-Application No.10-2023-0123693, Sep., 2023.
- [P1] Video Anomaly Detection Apparatus and Method using Relational Embedding. KR-Application No.10-2022-0156968, Nov., 2022.

PROJECTS

Research on Robust Neural Rendering-Based Large-Scale 3D Ultra-Precision Virtual Space Generation and Spatial Registration for Low-Quality Noisy Data DEEP LEARNING RESEARCHER • Development of large scene reconstruction by 3D Gaussian Splatting.	National Research Foundation of Korea May. 2024 - Apr. 2027
Collaborative Perception and Intelligence Framework for Hyper-connected Interaction among Human and Intelligent Things DEEP LEARNING RESEARCHER • Project Leader • Development of scene perception models: human pose estimation, action recognition, multi-object tracking, and re-identification.	Korea Electronics Technology Institute Apr. 2024 - Dec. 2025
Development of Anti-spoofing Model for Face Recognition Based on RGB Camera DEEP LEARNING RESEARCHER • Development of face anti-spoofing model robust to various spoofing attack.	Samsung Electronics Aug. 2023 - Jul. 2024
Development of Mobile Palmprint Recognition Algorithm DEEP LEARNING RESEARCHER • Development of one-stage real-time mobile network, which includes keypoint detection and palmprint recognition. • Development of real-time Android demo application for palmprint recognition.	Samsung Electronics Aug. 2022 - Jul. 2023

Deep Learning-Based Initial Identification and Tracking System for Missing Persons in Heterogeneous CCTV Images

DEEP LEARNING RESEARCHER

- Development of real-time multi-object tracking algorithm robust to occluded person.

National Research Foundation of Korea

Oct. 2018 - Dec. 2022

Development of AI Multi-Object Tracking and Behavior Analysis Technology

DEEP LEARNING RESEARCHER

- Development of robust feature extractor for the object detection network.

Hanwha Techwin

Oct. 2020 - Oct. 2021

Professional Services

Journal Reviewer

- IEEE Transactions on Graphics (TOG) 2025
- International Journal of Computer Vision (IJCV) 2024
- IEEE Transactions on Multimedia (TMM) 2024
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) 2023
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS) 2023
- IEEE Transactions on Circuits and Systems for Video Technology (TCSVT) 2023

Conference Reviewer

- International Conference on Learning Representations (ICLR) 2027
- Association for the Advancement of Artificial Intelligence (AAAI) 2026, 2027
- IEEE/CVF Computer Vision and Pattern Recognition (CVPR) - Outstanding Reviewer (2025) 2025, 2026
- Conference on Neural Information Processing Systems (NeurIPS) 2024, 2025, 2026
- European Conference on Computer Vision (ECCV) 2024, 2026
- IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2024
- The British Machine Vision Conference (BMVC) 2026

Invited Talks

- "3D Neural View Synthesis from Degraded Images," NAVER AI Lab Jul. 2025
- "3D Neural View Synthesis from Degraded Images," Korean Electronics Technology Institute Jul. 2025
- "Leveraging Circle of Confusion for 3D Neural View Synthesis," Korean Photonics Technology Institute May. 2025

TEACHING EXPERIENCES

Understanding and Using AI

TEACHING ASSISTANT

Yonsei University
Spring 2022 - Fall 2025

Deep Learning Lab.

TEACHING ASSISTANT

Yonsei University
Spring 2023

Digital Logic Circuit

TEACHING ASSISTANT

Yonsei University
Fall 2021

SKILLS

Research and Development Stacks

Main Languages Python, C/C++, MATLAB, Kotlin

AI Agents Claude, Codex